Institute of Technology of Cambodia

Department of Civil Engineering

Reinforcement Concrete

Project 01

Lecture : Dr. Sambath KY

Name	**:**	**Team**
ID	**:**	**e2022—-**
Group	**:**	**I4-GCI-A1**

2025-2026

Note that: it just sample, maybe Makara is the one who design the cover.

Reinforcement Concrete

Project 01

We can delete this since it no need

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1 Introduction

This is Our Team Intro.

2 Team Organization

3 Meeting Minutes & Progress Reports

4 Architectural & Structural Assumptions

This is Our Team Assumption.

5 Design & Optimization Process

This is Design

6 Detailed Design Calculations

6.1 Load

For imposed loads on floors according to Table 6.1 – 6.4 in EN 1991-1-1:2002 [3]

Category	Sub-category / Use	q_k [kN/m²]	Q_k [kN]
Category A	Areas for domestic and residential activities		
	- Floors	1,5 to 2,0	2,0 to 3,0
	- Stairs	2,0 to 4,0	2,0 to 4,0
	- Balconies	2,5 to 4,0	2,0 to 3,0
Category B	Office areas	2,0 to 3,0	1,5 to 4,5
Category C	Areas where people may congregate		
C1	Areas with tables (e.g. schools, cafes, restaurants, reading rooms)	2,0 to 3,0	3,0 to 4,0
C2	Areas with fixed seats (e.g. theatres, cinemas, churches)	3,0 to 4,0	2,5 to 7,0 (4,0)
C3	Areas without obstacles for moving people (e.g. museums, entrance halls)	3,0 to 5,0	4,0 to 7,0
C4	Areas with possible physical activities (e.g. dance halls, gyms)	3,5 to 5,0	3,5 to 7,0
C5	Areas susceptible to large crowds (e.g. concert halls, sports halls with stands)	5,0 to 7,5	3,5 to 4,5
Category D	Shopping areas		
D1	Areas in general retail shops	4,0 to 5,0	3,5 to 7,0 (4,0)
D2	Areas in department stores	4,0 to 5,0	3,5 to 7,0

Category E	Storage and industrial activities (general storage, accessible areas)	7,5	7,0

Table 6.1: Imposed loads on floors base on EN 1991-1-1:2002

No.	Room / Area	Category (EN 1991-1-1)	q_k (kN/m2)	Justificatio
1	Receptionist/ Reception	B	3	Office area with desks and waiting
2	Workstation / Open office	B	3	Standard office space
3	Meeting Room	C1	3	Area with tables (meeting furniture)
4	President Office	B	3	Private executive office
5	Vice President Offices	B	3	Individual office rooms
6	Accounting	B	3	Administrative office ar
7	General Sale & Administration	B	3	Office desks and workst
8	Copy / Printer Area	B	3	Office equipment area
9	Storages	E1	5	Light storage (files, office supplies) - upper
10	Lift Lobby / Lift Area	C3	4	Circulation area withou fixed obstacles
11	Stair (Main Stair)	A	4	Office building stair (not residential)
12	Restroom (Men)	B	3	Sanitary facilities withi
13	Restroom (Women)	B	3	Sanitary facilities withi
14	—	—	—	—
15	—	—	—	—
16	—	—	—	—
17	Entrance	C3	4	Entrance hall / free circulation area
18	Emergency Stair	A	4	Escape stair in office bu
19	Server& Technical Room	E1 (conservative)	5	IT equipment / racks – treated as light
20	Kitchen / Pantry	C1	3	Small office kitchen wit
21	Library	C3	4	Reading area without d (if heavy shelving → E1
22	Share Space	C3	4	Open collaborative area

7 Design Drawings

This is Our Team Design Drawings.

8 Results, Discussion & Conclusion

This is Our Team Conclusion.

References

[1] BSI, *BS EN 1992-1-1:2023 - Eurocode 2: Design of concrete structures - Part 1-1: General rules and rules for buildings, bridges and civil engineering structures*, BSI Standards Publication, 2023.

[2] BSI, *BS EN 1991-1-1:2025 - Eurocode 1: Actions on structures - Part 1-1: Specific weight of materials, self-weight of construction works and imposed loads for buildings*, BSI Standards Publication, 2025.

[3] British Standards Institution, *BS EN 1991-1-1:2002 - Eurocode 1: Actions on structures - Part 1-1: General actions - Densities, self-weight, imposed loads for buildings*, Incorporating corrigenda December 2004 and March 2009, BSI, 2002.

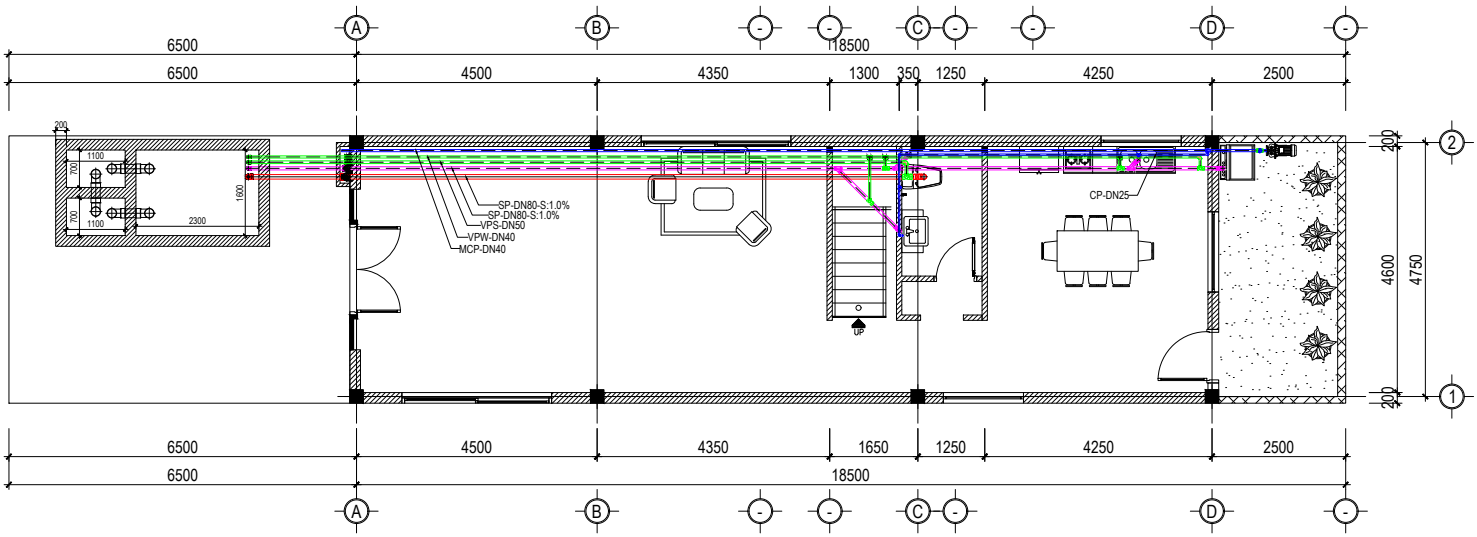
[4] CEN, *EN 1991-1-4:2005+A1 - Eurocode 1: Actions on structures - Part 1-4: General actions - Wind actions*, European Standard, Incorporating corrigendum January 2010, April 2010.

[5] CEN, *Eurocode 1: Actions on structures - Part 1-4: General actions - Wind actions*, EN 1991-1-4:2005+A1, English version, April 2010.

Note that : We may use the referennces provide by professor (check in telegram group) if it still not enought we can add more maybe.

Appendices

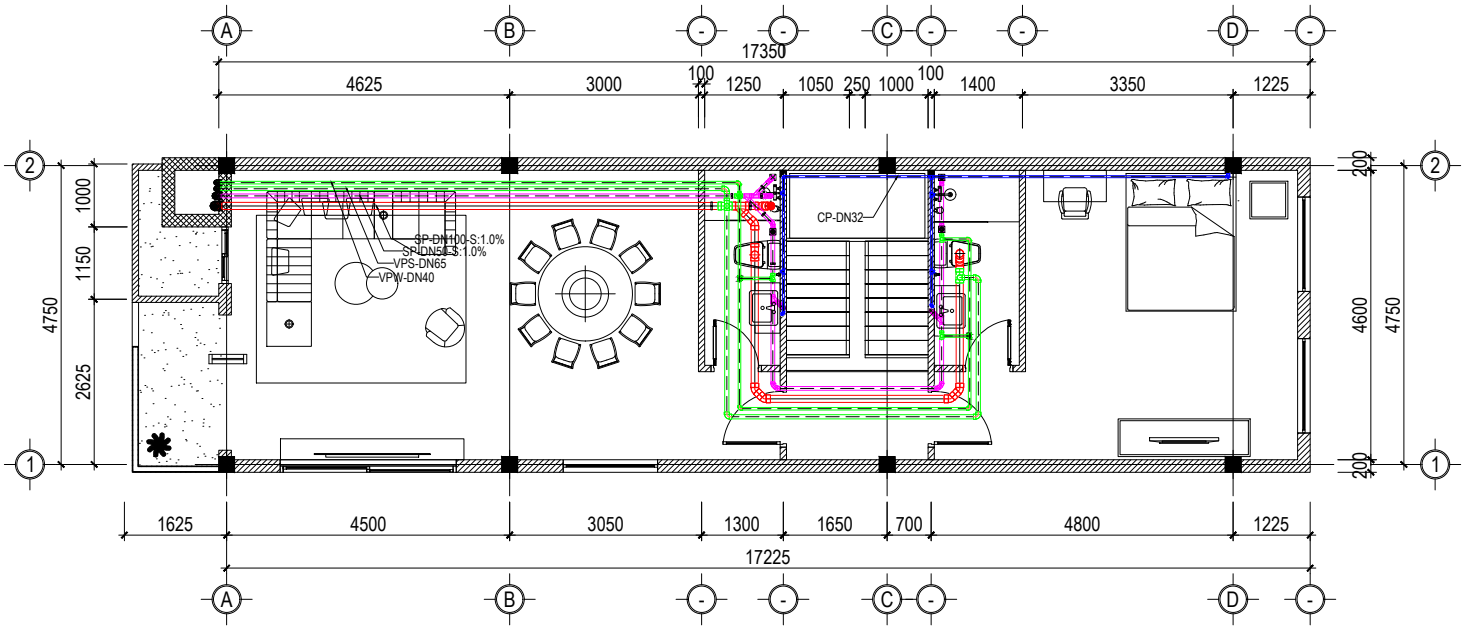
This is appendices



GROUND FLOOR
SCALE : 1/100

GROUND FLOOR		
SP	3"	80mm
WP	2"	50mm
VSP	2"	50mm
VWP	1.5"	40mm
CP	1.5"	25mm

Course: Installation and Sanatary Lecturer: MAY Raksmey	Student: CHUCH Manet ID: e20220847	Plumbing Plan



FIRST FLOOR
SCALE : 1/85

1ST FLOOR		
SP	4"	100mm
WP	2"	50mm
VSP	2.5"	65mm
VWP	1.5"	40mm
CP	1.5"	32mm

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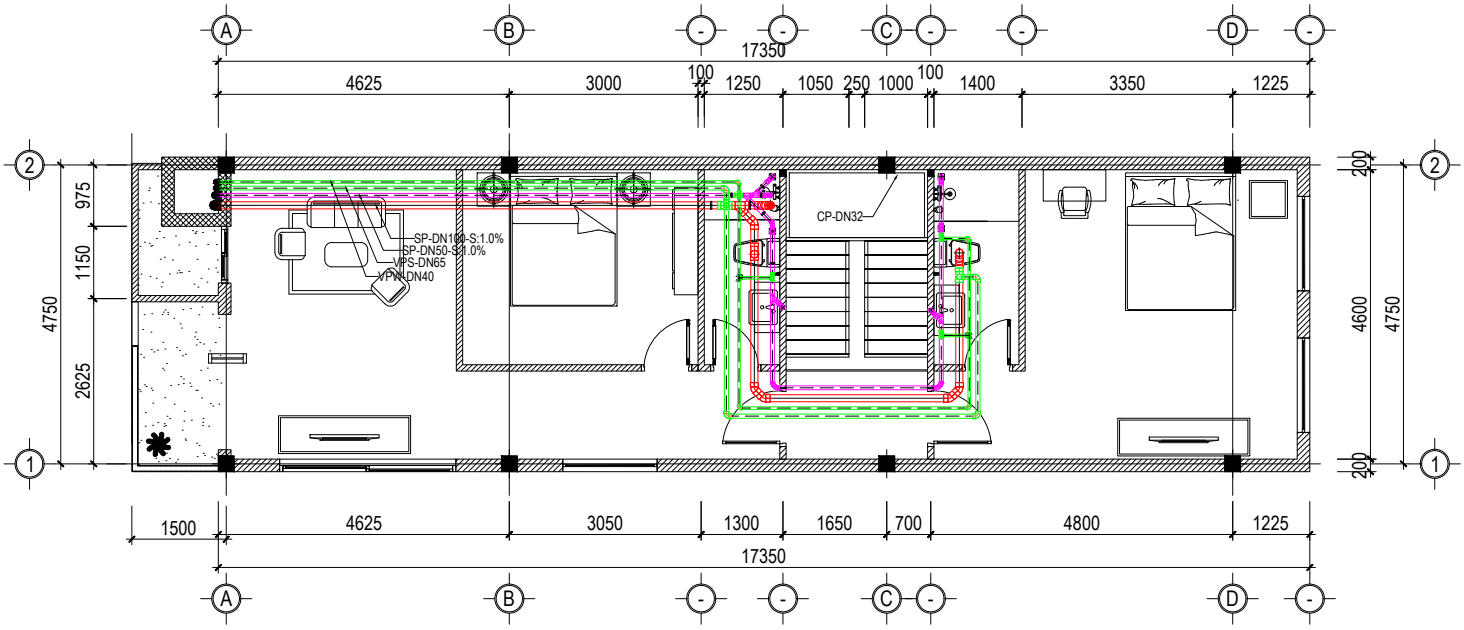

SECOND FLOOR
SCALE : 1/85

2ND FLOOR		
SP	4"	100mm
WP	2"	50mm
VSP	2.5"	65mm
VWP	1.5"	40mm
CP	1.5"	32mm

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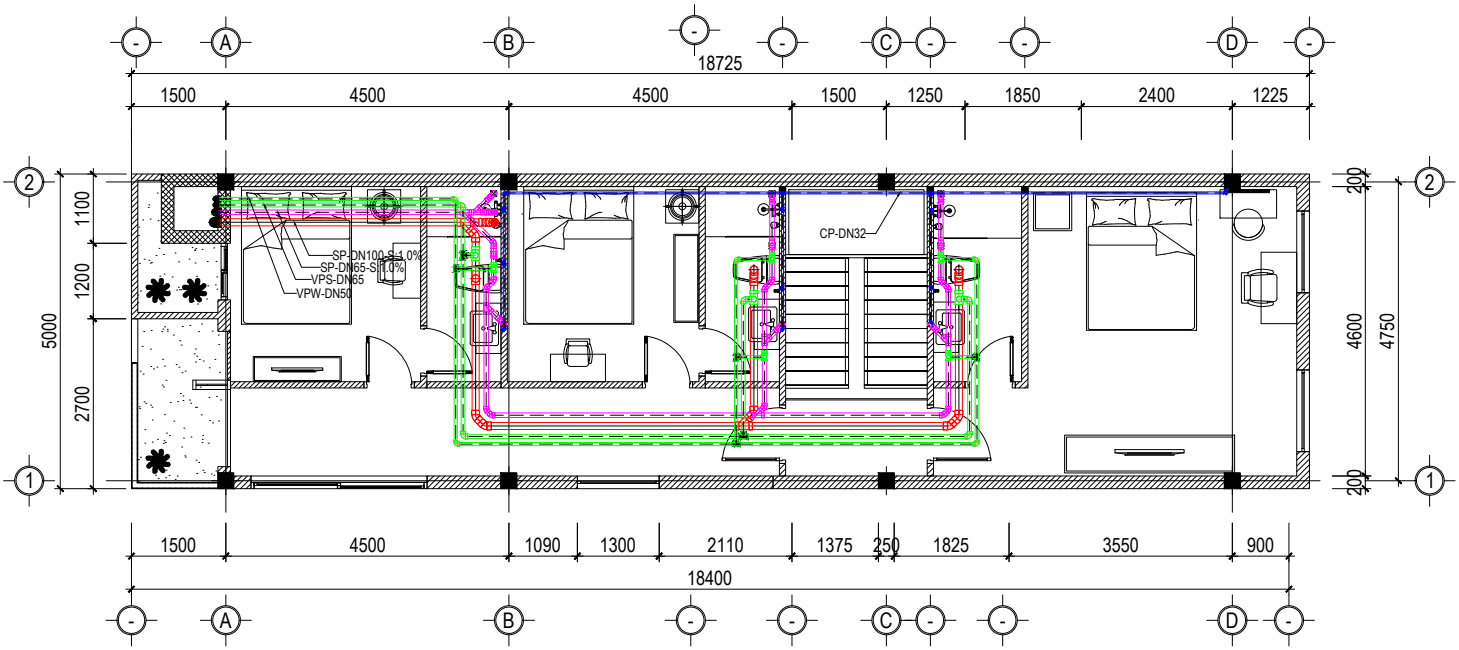

THIRD FLOOR
SCALE : 1/85

3RD FLOOR		
SP	4"	100mm
WP	2.5"	65mm
VSP	2.5"	65mm
VWP	2"	50mm
CP	1.5"	32mm

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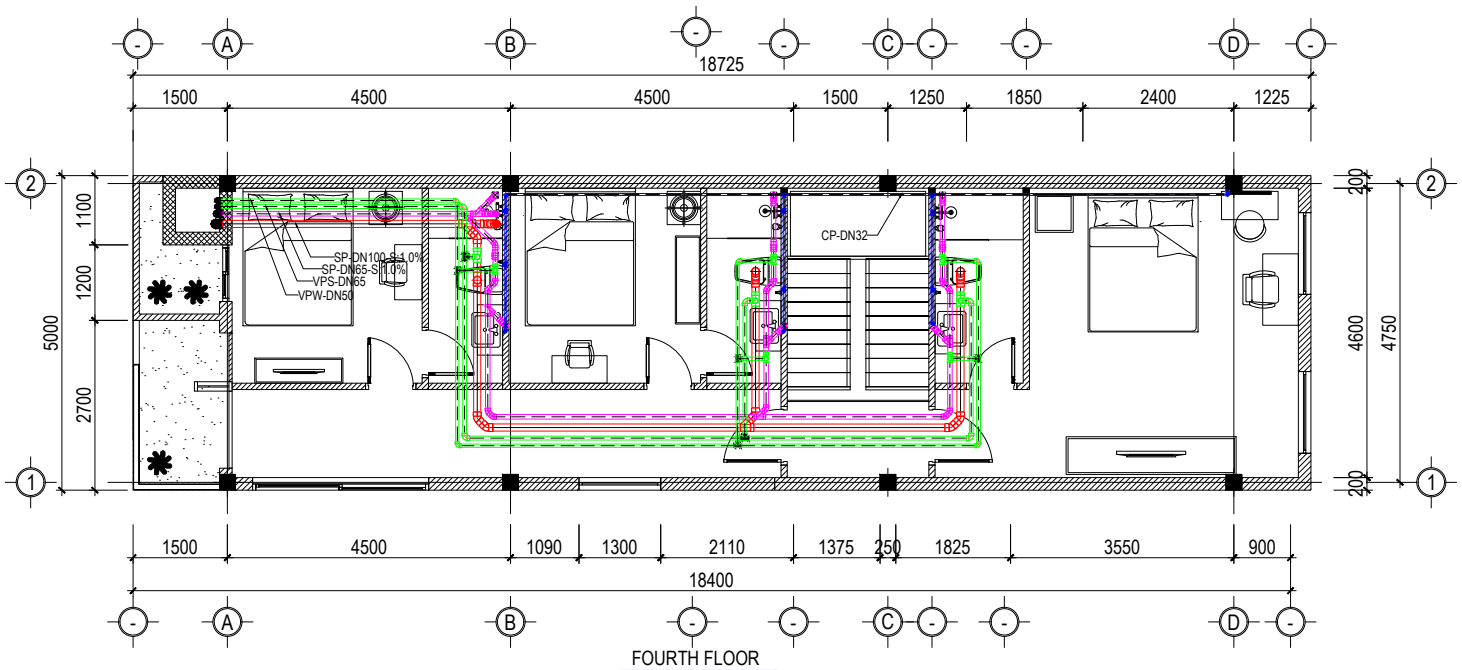

FOURTH FLOOR

SCALE : 1/85

3RD FLOOR		
SP	4"	100mm
WP	2.5"	65mm
VSP	2.5"	65mm
VWP	2"	50mm
CP	1.5"	32mm

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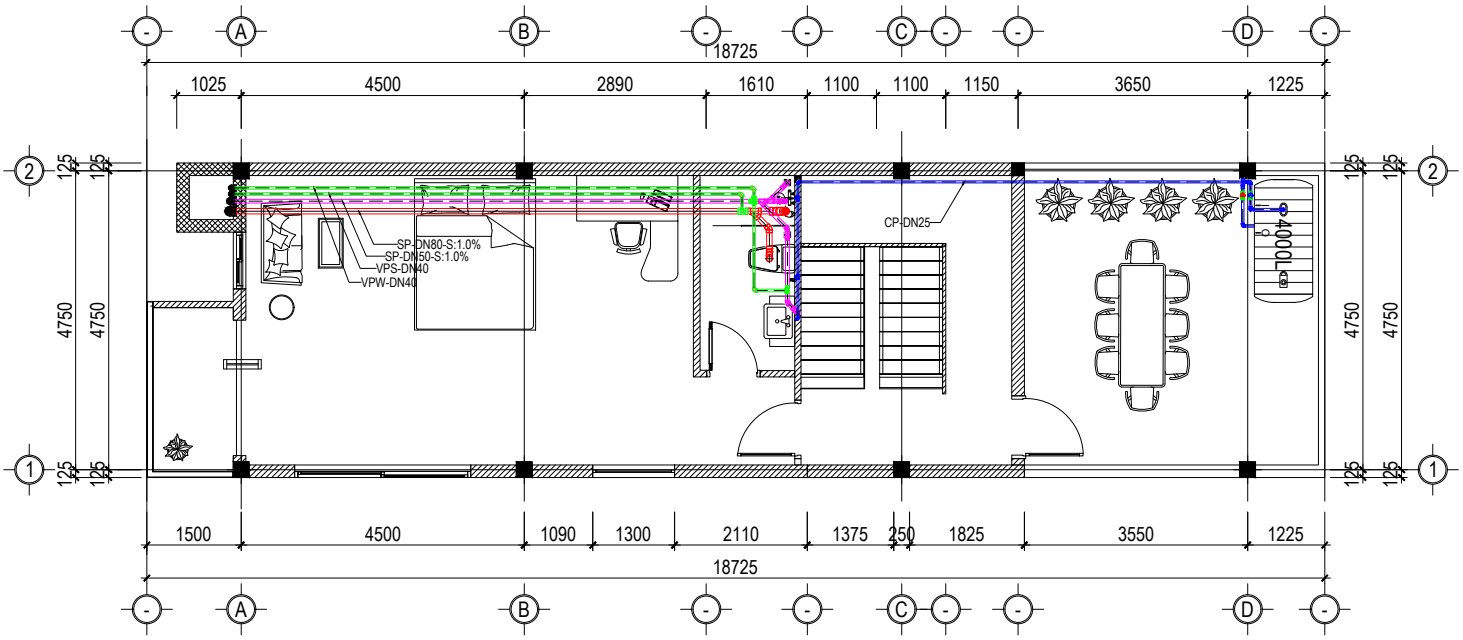

TERRACE FLOOR

SCALE : 1/85

GROUND FLOOR		
SP	3"	80mm
WP	2"	50mm
VSP	1.5"	40mm
VWP	1.5"	40mm
CP	1.5"	25mm

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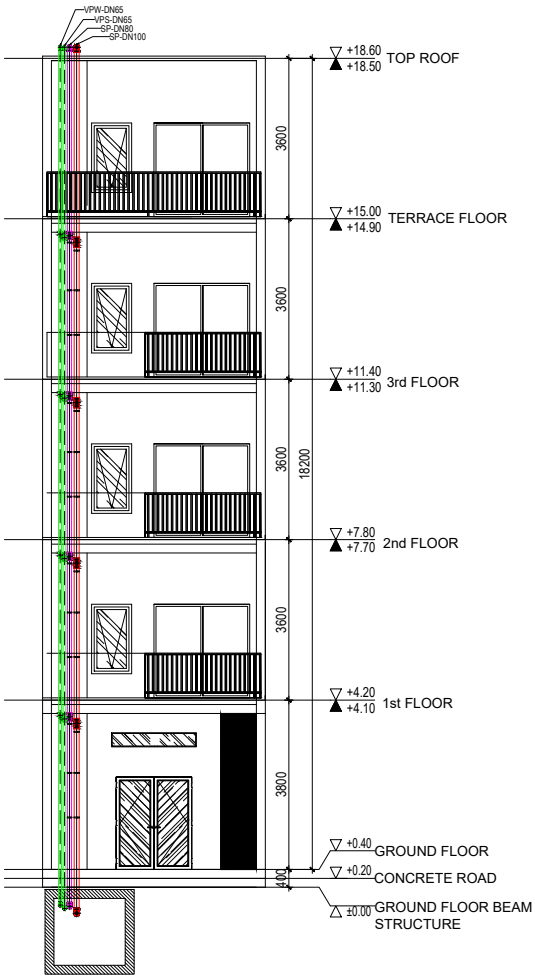

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